3. Given the function $f : x \rightarrow x^2 - 5x + 6$, evaluate

(i) $f(-1)$ (ii) $f(2a)$

(iii) $f(2)$ (iv) $f\left(x-1\right)$

4. A graph representing the function $f(x)$ is given in Fig.1.16 it is clear that $f(9) = 2$.

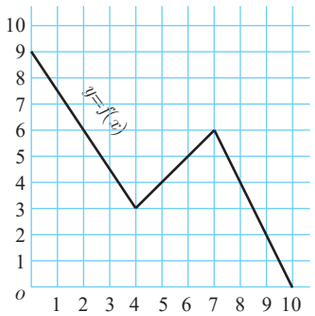


Fig. 1.16

(i) Find the following values of the function

(a) $f(0)$ (b) $f(7)$ (c) $f(2)$ (d) $f(10)$

(ii) For what value of x is $f(x) = 1$?

(iii) Describe the following (i) Domain (ii) Range.

(iv) What is the image of 6 under f?

5. Let $f(x) = 2x+5$. If $x \neq 0$ then find $\dfrac{f(x+2)-f(2)}{x}$.

6. A function f is defined by $f(x) = 2x - 3$

(i) find $\dfrac{f(0)+f(1)}{2}$.

(ii) find x such that $f(x) = 0$.

(iii) find x such that $f(x) = x$.

(iv) find x such that $f(x) = f(1-x)$.

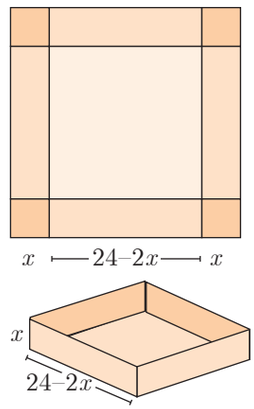


Fig. 1.17

7. An open box is to be made from a square piece of material, 24 cm on a side, by cutting equal squares from the corners and turning up the sides as shown (Fig.1.17). Express the volume V of the box as a function of x.

8. A function f is defined by $f(x) = 3 - 2x$. Find x such that $f(x^2) = (f(x))^2$.

9. A plane is flying at a speed of 500 km per hour. Express the distance 'd' travelled by the plane as function of time t in hours.

10. The data in the adjacent table depicts the length of a person forehand and their corresponding height. Based on this data, a student finds a relationship between the height (y) and the forehand length(x) as $y = ax + b$, where a, b are constants.

Length 'x' of forehand (in cm)	Height 'y' (in inches)
35	56
45	65
50	69.5
55	74

(i) Check if this relation is a function.

(ii) Find a and b.

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